

5

Moment of a Force

In this chapter, we will learn about

- the definition of the moment of a force
- the use of moments and couples
- the conditions for equilibrium of a rigid body
- interpreting and determining the centre of gravity

5.1 The turning effect of a force

In Chapter 2, we have discussed that a force can move an object from one place to another. Furthermore, it can also give a turning effect to a **rigid body**, i.e. an object with a fixed shape and size (Fig 5.1a). In this section, we will learn about the turning effect of forces and discuss its uses.

Flipped classroom
Watch this and answer the questions.

(i) (ii) (iii)

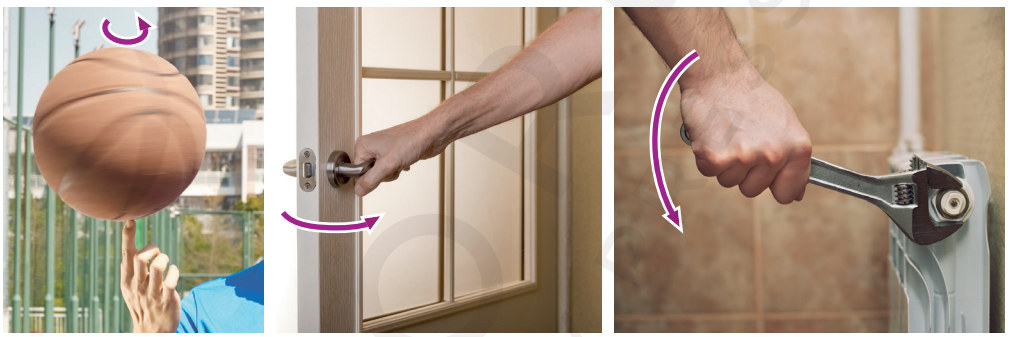


Fig 5.1a *Turning effect of a force: (i) spinning a ball, (ii) opening a door, (iii) tightening a nut by a spanner.*

DSE goal
Define moment of a force.

1 Moments

The turning effect of a force F about a certain point O is measured by the **moment** (or **torque**). It is defined as the product of the force F and its **perpendicular distance** $d_\perp$ from that point (Fig 5.1b). It is represented by the symbol τ and its unit is the **newton-metre** (**N m**).

moment about a point (N m) $\tau = Fd_\perp$ perpendicular **distance** of the force from that point (m)

force (N)

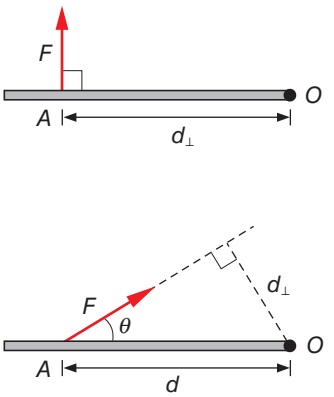


Fig 5.1b *Locating the perpendicular distance $d_\perp$.*

The perpendicular distance is the shortest distance between the point and the straight line which the force lies on. ▶

rigid body 剛體　　moment 力矩　　torque 轉矩

The moments about O are the same in the following two cases.

If F acts at point A and it makes an angle θ with AO (Fig 5.1c), $d_\perp = d \sin \theta$. Its moment about O can be further expressed as

$$\tau = Fd_\perp = Fd \sin \theta$$

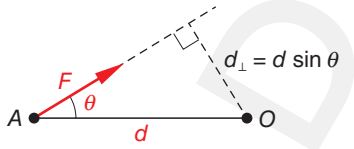


Fig 5.1c $\tau = Fd \sin \theta$.

This equation agrees with your daily-life experience.

1 The larger the force ($F\uparrow$), the larger the moment ($\tau\uparrow$) (Fig 5.1d).

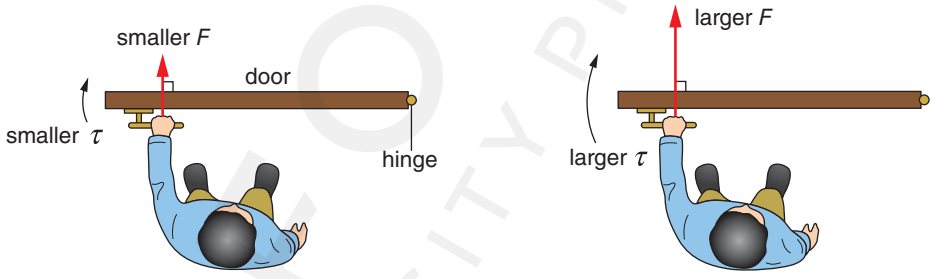


Fig 5.1d *A larger force produces a larger moment.*

2 The larger the perpendicular distance between the force and a certain point ($d_\perp\uparrow$), the larger the moment about this point ($\tau\uparrow$) (Fig 5.1e).

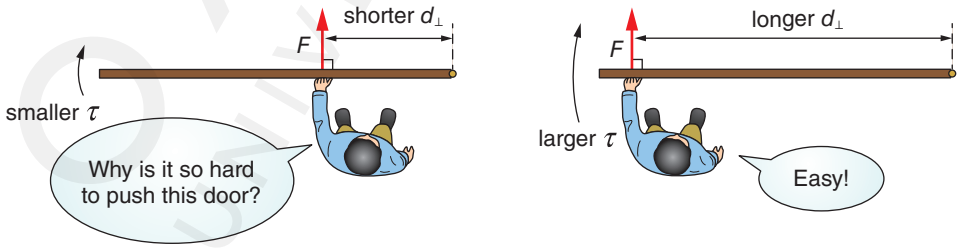
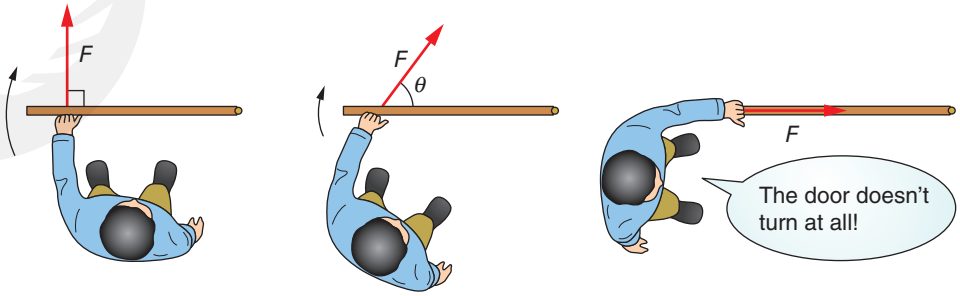


Fig 5.1e *The position at which a force is applied affects the moment.*

3 A force acting at a certain point provides the largest moment when $\sin \theta = 1$, i.e. when $\theta = 90°$. It provides the smallest moment when $\sin \theta = 0$, i.e. when $\theta = 0°$ (Fig 5.1f).



(i) Largest τ. *(ii) Smaller τ.* *(iii) Smallest τ ($\tau = 0$).*

Fig 5.1f *The relationship between τ and θ.*

→ This is a Video Quiz about the experiment 'Turning effect of a force'.

For a force F acting at point A, its moment about point O can also be defined as the product of the **component of F perpendicular to AO** (Fig 5.1g), denoted by $F_\perp$, and the distance d between A and O. Only $F_\perp$ produces the turning effect about O.

$$\tau = F_\perp d$$

As $F_\perp = F \sin\theta$, the above can be further expressed as

$$\tau = F_\perp d$$
$$= F \sin\theta \times d = Fd \sin\theta$$

This agrees with the equation obtained on p.197.

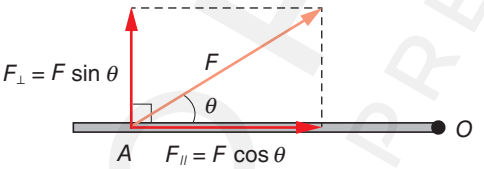
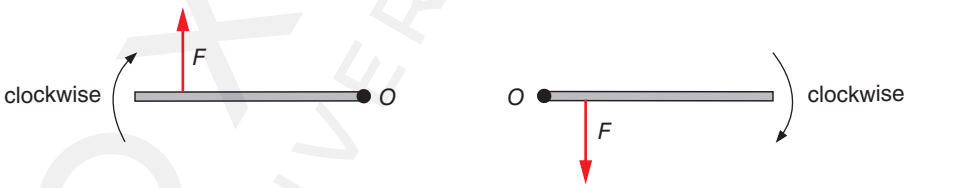


Fig 5.1g *Resolving F into components parallel and perpendicular to AO.*

Moment is considered a vector. ▶ Moment has direction. For the turning effect on a plane, the direction of moment can be described in terms of **clockwise** or **anticlockwise** (Fig 5.1h).

(i) Examples of clockwise moment



(ii) Examples of anticlockwise moment

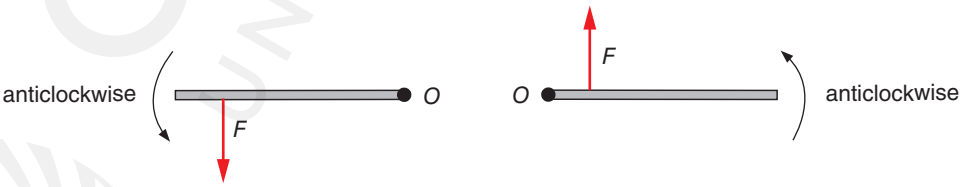
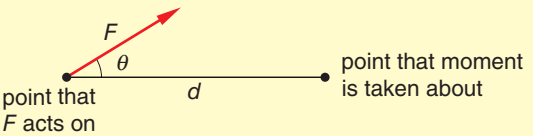


Fig 5.1h *Directions of moment about point O.*

The following two steps are usually used in finding moments.

1 Find the magnitude by using $\tau = Fd \sin\theta$.



Depending on the information given, $\tau = F_\perp d$ or $\tau = Fd_\perp$ may be used in some cases.

2 Determine the direction of the moment.

The above discussion applies to objects of any shape (Fig 5.1i).

An object may turn about a certain point when it is acted on by a moment. This point is called the **pivot** and is usually fixed in position. In the case of pushing a door, the hinges of the door are the pivot (Fig 5.1j).

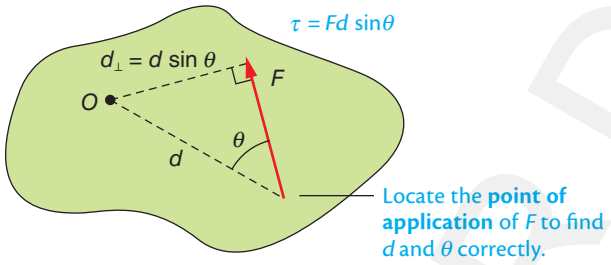


Fig 5.1i *F acting on an object of irregular shape.*

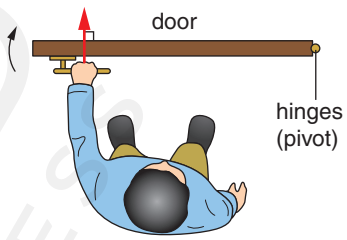
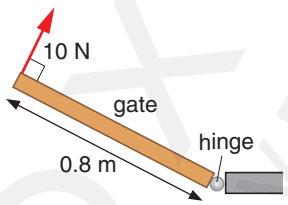


Fig 5.1j *An example of pivot.*

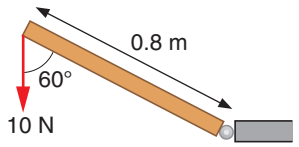
Example 1 Moment in pulling a gate

A boy pulls a gate with a force of 10 N in different ways as shown in the following top-view diagrams. Find the moment of the force about the hinge in each case.

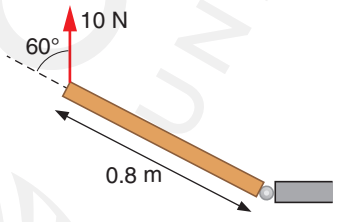
(a)



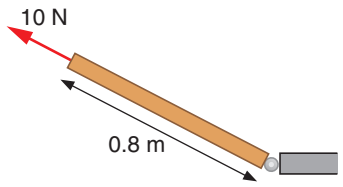
(b)



(c)



(d)



Solution

Apply $\tau = Fd \sin\theta$.

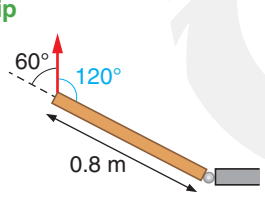
(a) Moment = $10 \times 0.8 \times \sin 90° = 8$ N m (clockwise)

(b) Moment = $10 \times 0.8 \times \sin 60° = 6.93$ N m (anticlockwise)

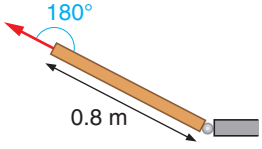
(c) Moment = $10 \times 0.8 \times \sin 120° = 6.93$ N m (clockwise)

(d) Moment = $10 \times 0.8 \times \sin 180° = 0$

Tip



Tip



pivot 支點

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 A force acts at point P as shown (Fig a).

What is the moment of this force about point O?

A $10 \times 0.2 \times \sin 60°$

B $10 \times 0.2 \times \cos 60°$

C $10 \times 0.3 \times \sin 50°$

D $10 \times 0.3 \times \cos 50°$

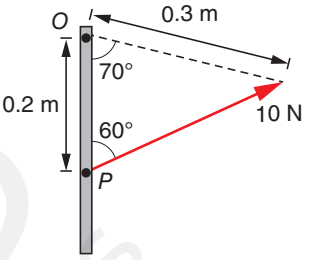


Fig a

2 A worker uses a puller to remove a nail. Figure b shows all the forces acting on the puller. Determine the direction of the moments about the pivot due to different forces (Table a).

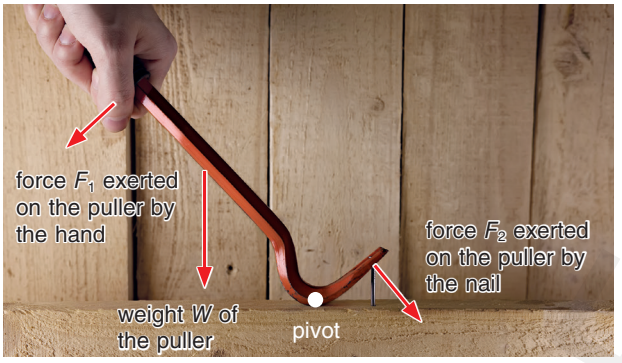


Fig b

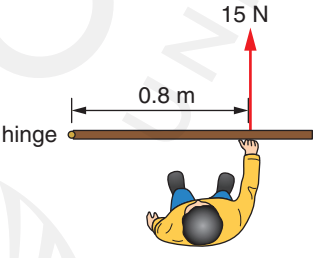
Force	Direction of moment
F_1	clockwise / anticlockwise
F_2	clockwise / anticlockwise
W	clockwise / anticlockwise

Table a

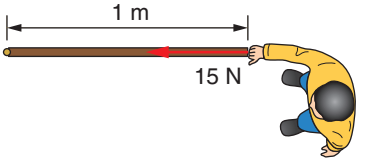
Learn by practice 1

1 Mickey is pushing a door in the following four ways. Find the moment due to the pushing force about the hinge in each case.

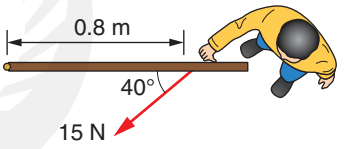
(a)



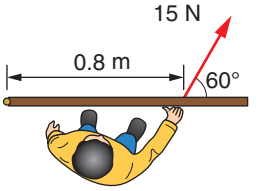
(b)



(c)



(d)



2 F_1 and F_2 act on a square box as shown (Fig a). The length of its side is 0.2 m. O is the centre of the box. It is known that F_1 = 12 N and F_2 = 15 N, find the moments of F_1 and F_2 about O.

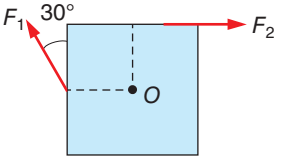


Fig a

2 Addition of moments

a Net moment

All moments acting on an object **about the same point** can be added to give the net moment. Signs should be assigned to the moments according to their directions. If the clockwise direction is taken as positive, the anticlockwise direction will become negative.

Example 2 | Net moment on a wheel

Children P and Q turn a wheel of radius 0.2 m. Find the net moment acting on the wheel about its centre O in each of the following cases.

(a) P pushes with 3 N upwards and Q pushes with 5 N downwards (Fig a)

(b) P pushes with 3 N upwards and Q pushes with 5 N upwards (Fig b)

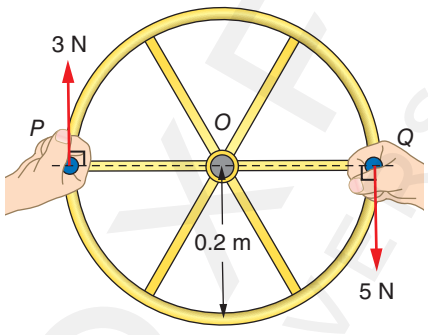


Fig a

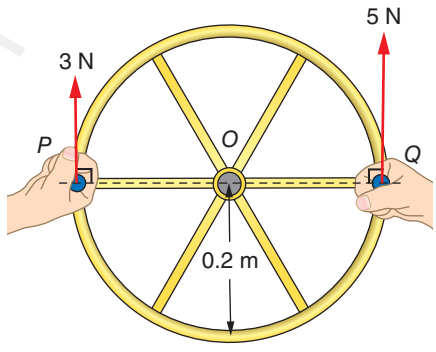


Fig b

Solution

Take the clockwise direction as positive. Apply $\tau = Fd_{\perp}$.

(a) $\tau_{net} = 3 \times 0.2 + 5 \times 0.2 = 1.6$ N m

The net moment is 1.6 N m (clockwise).

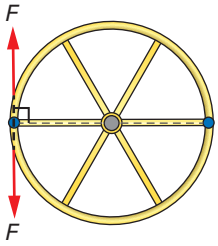
(b) $\tau_{net} = 3 \times 0.2 - 5 \times 0.2 = -0.4$ N m

The net moment is 0.4 N m (anticlockwise)

We simplified the discussion in Ch 3 and 4 by assuming the size of the object to be negligible. For an object whose size is not negligible, the relationship between net force and acceleration should be modified as follows:
if the net force acting on the object is zero, the acceleration of the centre of gravity of the object is zero.

A couple, which has a net force of zero, only turns an object without any acceleration of the centre of gravity. This does not violate Newton's second law of motion.

If two equal and opposite forces act in the same line, they do not form a couple and the resultant moment is zero.



b Couple

▶ In Figure 5.1k on p.202, two **equal and opposite** forces act on a steering wheel and **do not act in the same line**. We say that the two forces form a **couple**. The resultant moment due to a couple is not zero, so a couple can turn an object. However, the resultant force of a couple is zero, so a couple does not move an object from one place to another.

couple 力偶



Fig 5.1k *Two equal and opposite forces act on a steering wheel in different lines.*

Note that the forces in the situations below do not form a couple (Fig 5.1l).

When you ride a bicycle, you step on the pedals. Since the forces acting on the pedals by both legs point downwards and do not act at the same time, they do not form a couple.





When children are playing on a see-saw, the forces acting on the see-saw by both children point downwards, so they do not form a couple.

Fig 5.1l *These forces do not form couples.*

Know more

Moment of a couple

Consider a cyclist turning a handlebar (Fig a). The forces that the hands exert on the handlebar form a couple. Figure b shows how the moment of the couple about *O* can be found.

In general, for two forces *F* with perpendicular distance *D* forming a couple, the magnitude of the moment of the couple can be expressed as

$$\tau_{couple} = F \times D$$

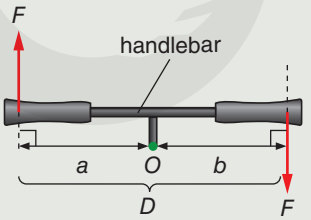
It can be further proven that this magnitude is the same for all points in the plane of the couple (Fig c).



Fig a

If each force of a couple makes an angle θ to the handlebar, the moment of the couple is

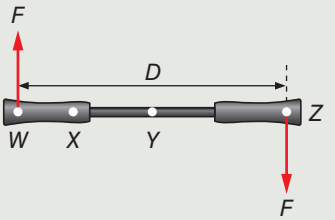
$\tau = FD \sin \theta$



Moment about $O = F \times a + F \times b$

$= F(a + b)$

Fig b



The moments about *W*, *X*, *Y*, and *Z* are all equal to *FD*.

Fig c

DSE goal

Discuss the uses of torques and couples.

3 Uses of moments and couples

Moments (Fig 5.1m) and couples (Fig 5.1n) are commonly found in daily life.

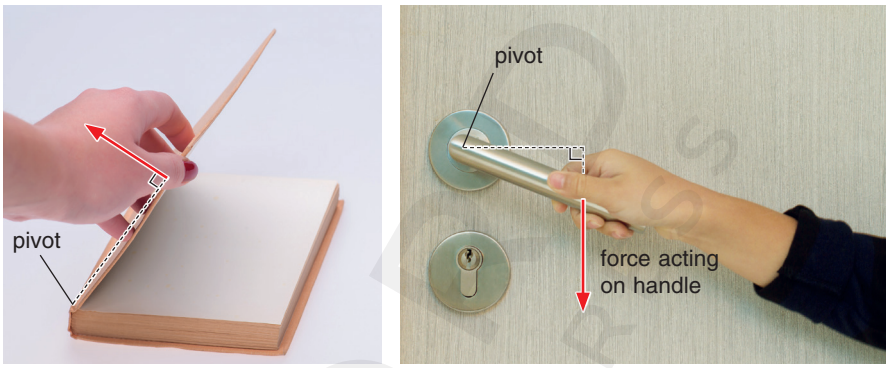


Fig 5.1m *Moments are used in opening a book and turning a handle.*

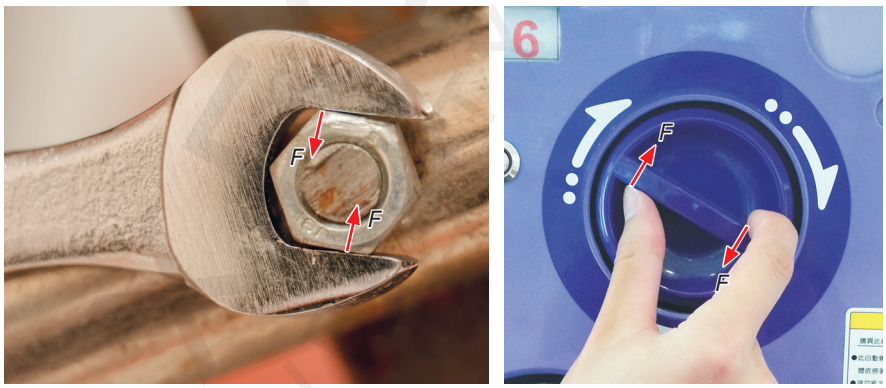


Fig 5.1n *Couples are used in turning a nut and turning a knob.*

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Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 *A* and *B* push a rotator from rest as shown (Fig a).

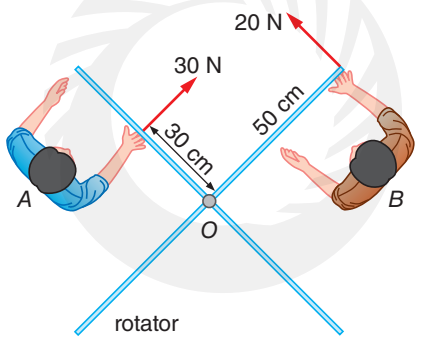


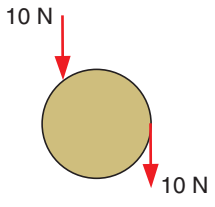
Fig a

(a) Find the net moment about *O*.

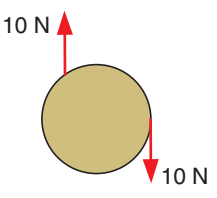
(b) In which direction will the rotator turn?

2 Two forces act on a circular object. In which of the following cases do the forces form a couple?

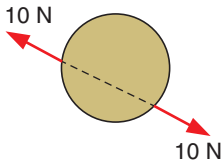
A



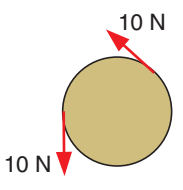
B



C



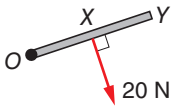
D



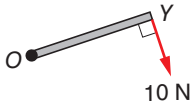
Exercise 5.1

1 In which of the following diagrams is the moment of the force about O the largest? Given: X is the mid-point of OY.

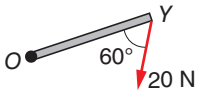
A



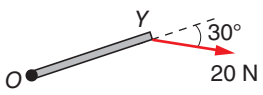
B



C

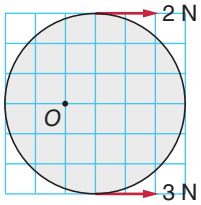


D

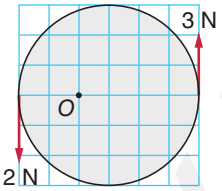


***2** Two forces are acting on a rigid body. In which of the following figures does the net moment about point O have the largest magnitude?

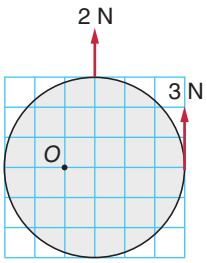
A



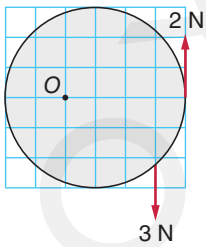
B



C



D



3 A ball is connected to point X by a light rod of length 0.7 m (Fig a). The magnitude of the moment due to its weight about X is 17.5 N m. Find its weight.

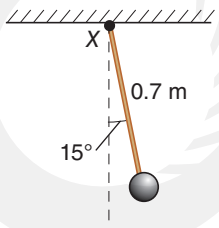


Fig a

4 A hand applies two forces F of the same magnitude on a knob of radius 2.5 cm (Fig b). The two forces form a couple and produce a moment with magnitude 0.1 N m. Find the magnitude of each force.

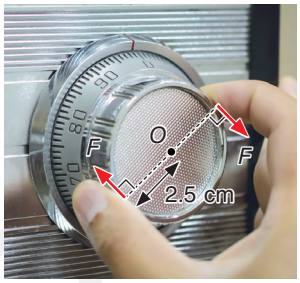


Fig b

***5** A metal can is tightly closed. Patrick uses a key to get the lid off (Fig c). If he uses a longer tool, such as a screw driver, how will the force F needed to open the can change? Explain briefly.

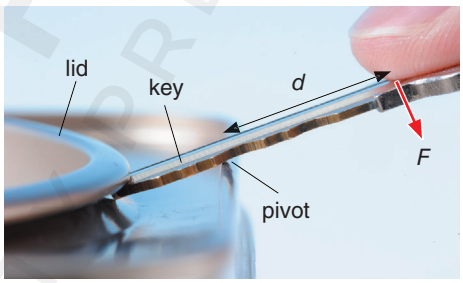


Fig c

***6** Two 5-N forces P and Q act on a square block (Fig d). C is the centre of the block and OC = 1 m.

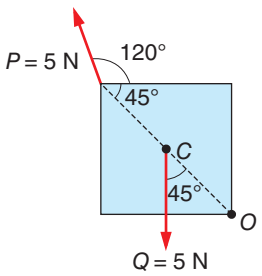


Fig d

(a) Find the moment of each force about O.

(b) Hence find the net moment about O.

(c) Find the net force along the vertical direction.

***7** In Figure e, Nick holds the thick end of a baseball bat while Samuel holds the thin end. They try to turn the bat in opposite directions. Who do you think can turn the bat if they apply forces of the same magnitude on it? Explain your answer.

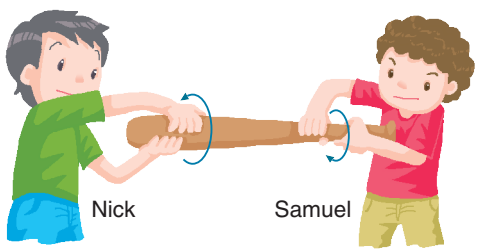
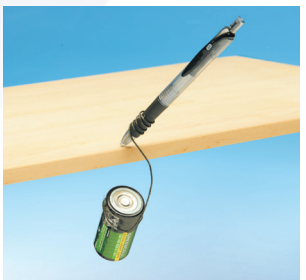


Fig e

5.2 Equilibrium of a rigid body

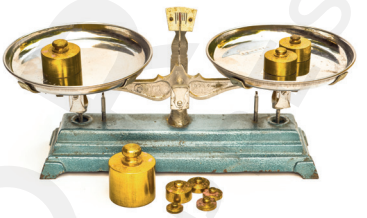
A rigid body remaining at rest (not moving nor rotating) is in **equilibrium**. People work with equilibrium in many activities such as sports, measurements, and games (Fig 5.2a). In this section, we will learn about the conditions for equilibrium.

A pen and a dry cell can be supported and stay in equilibrium in a weird form as shown. Why is it the case?





(i) Performing human flag.



(ii) Measuring mass using a beam balance



(iii) Playing stacking game.

Fig 5.2a *Applications of equilibrium.*

→ This simulation allows Ss to control the size and positions of the applied forces. Ss can also change the position of the pivot. It is available on OUP web in PC version (.exe) only.

DSE goal

State the conditions for equilibrium of forces acting on a rigid body.

1 Conditions for equilibrium

→ This is a Video Quiz about this experiment.

We shall do an experiment to investigate the conditions for an object to be in equilibrium.

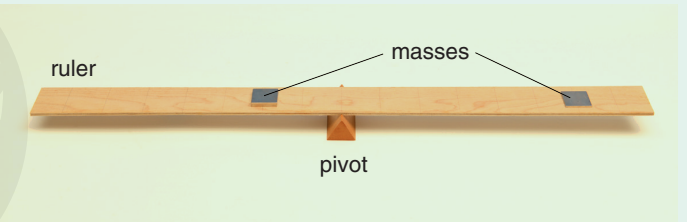
Simulation Video

→ This video demonstrates Expt 5a.

Experiment 5a Moments on a balanced ruler

1 Balance a ruler on a pivot.

2 Put two different masses on the ruler such that the ruler is balanced.



3 Calculate the clockwise and anticlockwise moments about the pivot.

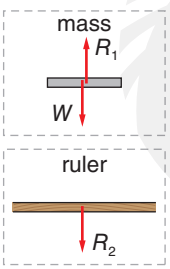
4 Repeat steps 1 to 3 using different masses.

Discussion

Is the sum of clockwise moments always equal to the sum of anticlockwise moments (i.e. net moment = 0) when the ruler is balanced?

The force acting on the ruler by a mass is equal to the weight of the mass.

Proof



The mass is at rest: $R_1 = W$

Action-and-reaction pair: $R_2 = R_1$

$\therefore R_2 = W$

equilibrium 平衡

Know more

The working principle of a beam balance

Do you know the working principle of a beam balance?



▸ p.227

In the experiment, when the ruler is in equilibrium, the net moment about the pivot is zero. Indeed, there are two conditions for a rigid body to be in equilibrium:

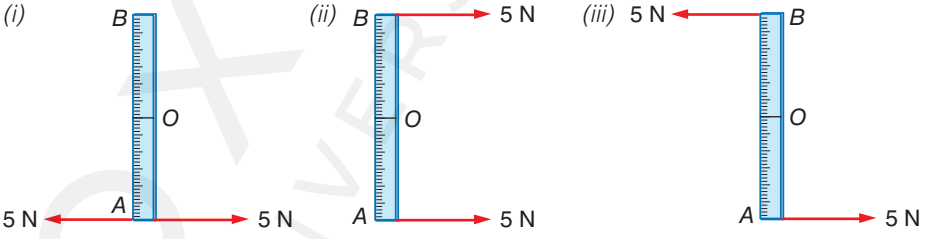
1 **Net force = 0**

2 **Net moment about any point = 0, i.e.**
sum of clockwise moments = sum of anticlockwise moments

When an object is in equilibrium, the net moment about **any point** on the object is always zero. By contrast, when an object is not in equilibrium, the net moments about certain points are non-zero. This is illustrated in the following example.

Example 3 Which ruler is in equilibrium?

A light ruler AB which is 50 cm long is initially at rest on a smooth horizontal table. O is the mid-point of the ruler. The following top-view diagrams show two forces acting on the ruler in three different ways.



(a) Find the net force and the net moment acting on the ruler about O and A in each case. Take the direction towards the right and the clockwise direction as positive.

(b) The ruler in one of the above cases is in equilibrium. Which case is it?

Solution

(a)

	(i)	(ii)	(iii)
Net force	$F_{net} = 5 - 5 = 0$	$F_{net} = 5 + 5 = 10$ N	$F_{net} = 5 - 5 = 0$
Net moment about O	$\tau_{net} = 5 \times 0.25 - 5 \times 0.25 = 0$	$\tau_{net} = 5 \times 0.25 - 5 \times 0.25 = 0$	$\tau_{net} = -5 \times 0.25 - 5 \times 0.25 = -2.5$ N m
Net moment about A	$\tau_{net} = 0$	$\tau_{net} = 5 \times 0.5 = 2.5$ N m	$\tau_{net} = -5 \times 0.5 = -2.5$ N m

(b) Case (i)

The forces in (iii) form a couple. The perpendicular distance between the forces is 50 cm.
Magnitude of $\tau = 5 \times 0.5 = 2.5$ N m

The moment of a force about its point of application is zero since $d_\perp$ is zero.

Tip
There is no net force and no net moment about different points on the ruler in this case.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

(For Q1–2). Rocks P and Q are put on a light (i.e. with negligible mass) plank which is supported at O (Fig a). The weights of P and Q are W_P and W_Q.

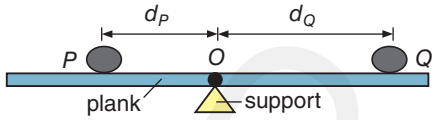


Fig a

1 In each of the following situations, the plank is **in equilibrium**. Complete the table below by considering the **net moment about O**.

	W_P / N	W_Q / N	d_P / m	d_Q / m
(a)	6		0.1	0.15
(b)		5	0.08	0.56
(c)	2	10	0.4	

2 The support exerts an upward force F on the plank. Find F in each situation in Q1 by considering the **net force along the vertical direction**.

2 Centre of gravity

DSE goal

Interpret the centre of gravity and determine it experimentally.

DSE exam

13(1A)Q14, 14(1A)Q3, 19(1A)Q4

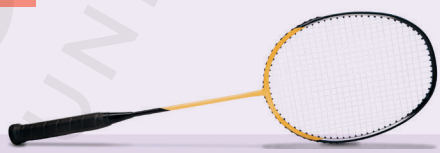
Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

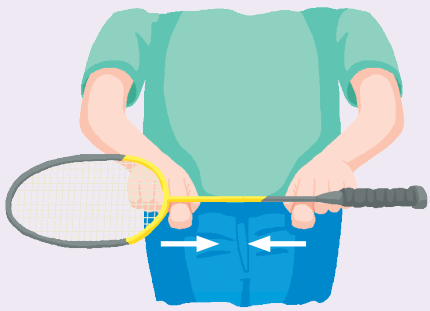
Supporting a racket using one finger

What you need



What to do

Support the racket horizontally using two fingers. Shift the fingers towards the centre slowly until they meet. Try to support the racket using one finger at the meeting point.



What you find

Can you support the racket with only one finger at this point?



Fig 5.2b *A turtle supported at one point only.*

The turtle shown in Figure 5.2b is in equilibrium when it is supported at one point only, just like the racket in **Hands-on activity**. To understand how this can be done, we should know what the **centre of gravity** (c.g.) is.

Every rigid body has a fixed point at which its weight seems to act. The point is called the centre of gravity.

The c.g. of a **uniform symmetrical** object, such as a rod of constant diameter and uniform density, is located at its **centre** (Fig 5.2c). The c.g. of a non-uniform object can be located experimentally, as illustrated in Experiment 5b.

The **Hands-on activity** on p.207 shows how the c.g. of a long object can be located.

In this part, *g* is presumably uniform in a body so that the centre of gravity and the centre of mass of the body are at the same position.

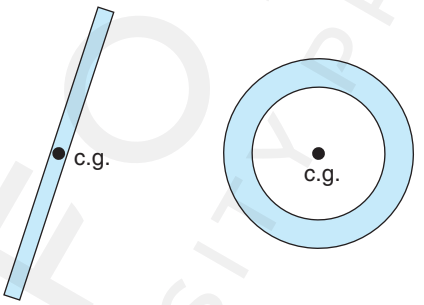


Fig 5.2c *The c.g. of uniform symmetrical objects.*

Video

→ This video demonstrates Expt 5b.

Experiment 5b Locating c.g.

1 Cut a random shape out of carboard.

2 Suspend the shape and a piece of string, which is tied to a mass, from a point near the edge with a pin (Fig a). The shape should be able to rotate freely about the pin. Mark the string's position on the shape (Fig b).

3 Repeat step 2 by using a different point. The c.g. of the shape is at the intersection of the two markings.

4 Support the shape at its c.g. with a pencil (Fig c).

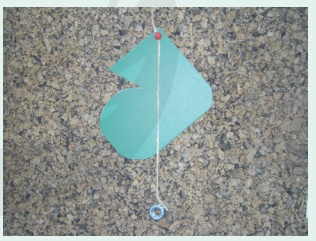


Fig a

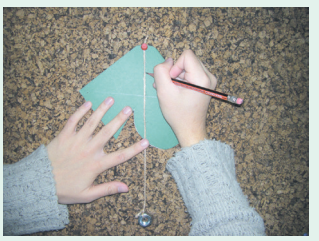


Fig b



Fig c

Discussion

Can you support the shape with the pencil?

centre of gravity 重心

There is friction between the pin and the shape. If the point of suspension is too close to the c.g., the moment due to the weight may be too small to overcome the effect of the friction.

In the above experiment, if the c.g. of the shape is not vertically under the point of suspension *X*, the moment due to the weight of the shape will cause the shape to rotate (Fig 5.2d). The shape can reach equilibrium only when its c.g. is vertically below *X*. Therefore, **the marking indicates all possible positions of the c.g.**

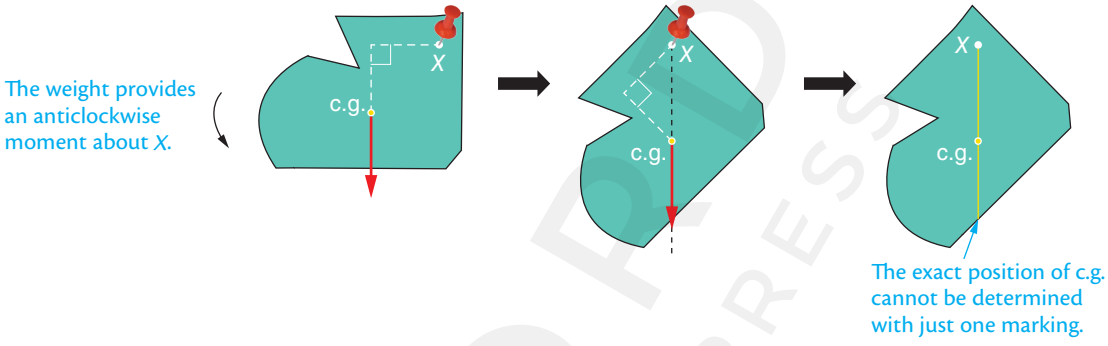


Fig 5.2d *The shape will rotate until X and c.g. are aligned vertically.*

When the shape is suspended at another point *Y*, the c.g. will also be vertically below *Y* when equilibrium is reached. This explains why the c.g. is at the intersection of the two markings (Fig 5.2e).

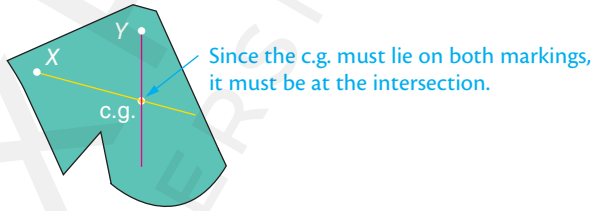


Fig 5.2e *Locating c.g. at the intersection of two markings.*

To keep an object stable by supporting it at a point, the supporting point and the c.g. must align vertically so that the weight of the object and the supporting force act along the same line (Fig 5.2f).

Can you now explain how the pen and the dry cell on p.205 stay in equilibrium?

Video

→ This video demonstrates a balancing trick using a pen and a dry cell.

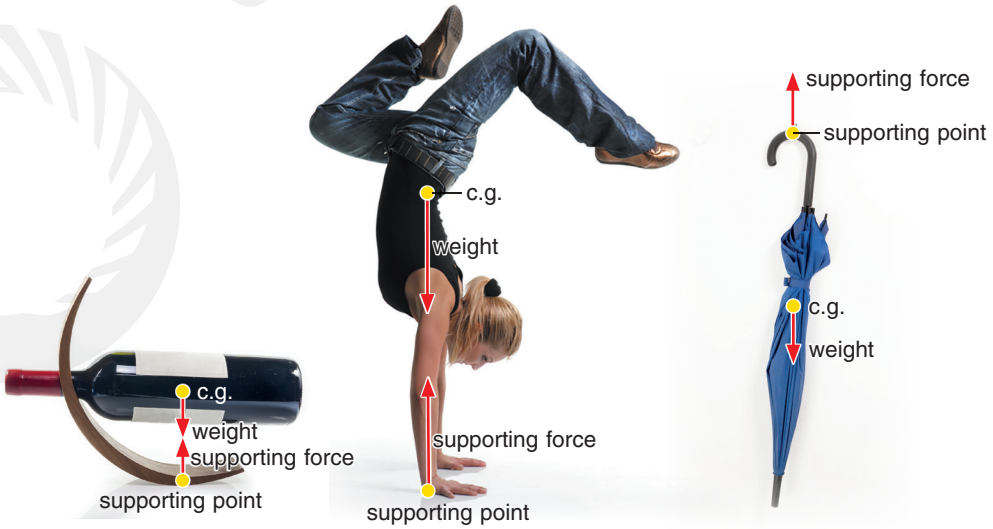


Fig 5.2f *Supporting point and c.g. align vertically.*

We can do a balancing trick by making use of the position of c.g.

Video

→ This video demonstrates the Hands-on activity.

Balancing trick: tilting can

Hands-on activity

What you need



What to do



1 Fill the empty can with about 100 ml of water.

2 Balance the can on its slanted edge at its bottom. You may need to adjust the amount of water inside to make it balance.

What you find

Where is the c.g. of the can when it is in equilibrium?

Know more

Position of c.g.

It is hard for an object to keep its balance if its c.g. is above the supporting point. In contrast, it is easy for an object to restore its balance if its c.g. is below the supporting point. Do you know why?

▸ p.228

Google Forms

Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* If an object is only acted on by its weight and an upward force, and the object is in equilibrium, the upward force and the weight must act along the same line. (T/F)

2 Which of the following statements about the centre of gravity (c.g.) of an object must be correct? Put a tick in the corresponding box(es).

- ☐ It is at the centre of a uniform circular object.
- ☐ It is at the centre of a rectangular object.
- ☐ It is within the body of the object.
- ☐ It is the point where all the mass of the object is located.
- ☐ It is the point where the weight of the object seems to act.

DSE goal

Solve problems involving a fixed pivot.

DSE exam

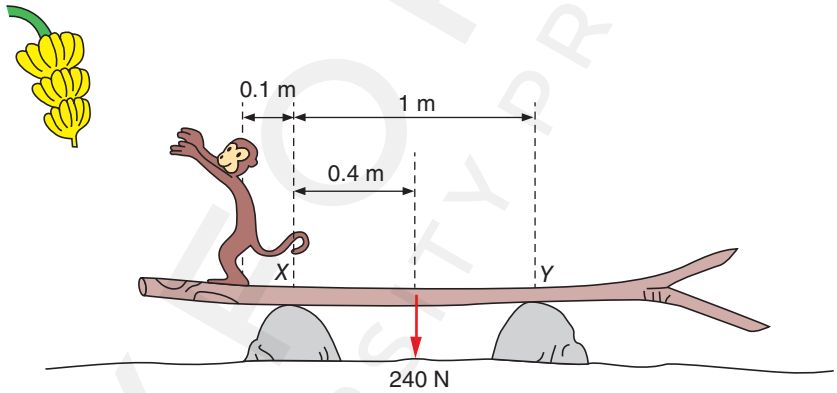
12(1A)Q6, 16(1A)Q11, 17(1A)Q8, 18(1A)Q6, 18(1B)Q5, 19(1A)Q7, 20(1A)Q7, 23(1A)Q12, etc

3 Analysing objects in equilibrium

We can now apply the conditions for equilibrium and analyse the forces acting on an object in equilibrium.

Example 4 Tree trunk in equilibrium

A monkey walks on a fallen tree trunk which is supported horizontally by two stones at X and Y. The stones are 1 m apart. The monkey's weight is 300 N. The trunk's weight is 240 N and its centre of gravity is 0.4 m from X. The monkey is 0.1 m to the left of X. The trunk remains at rest.

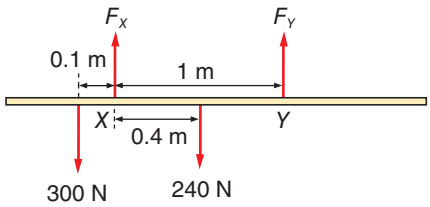


(a) The stones at X and Y exert vertical forces F_X and F_Y on the tree trunk. Find the magnitude of these two forces.

(b) The monkey walks to the left to pick some bananas. What is its maximum distance d from X without making the trunk topple?

Solution

(a) Free-body diagram for the tree trunk:



Take moment about Y.

Total clockwise moments = total anticlockwise moments

$$F_X \times 1 = 300 \times (0.1 + 1) + 240 \times (1 - 0.4)$$

$$F_X = 474 \text{ N}$$

Take moment about X.

Total clockwise moments = total anticlockwise moments

$$240 \times 0.4 = 300 \times 0.1 + F_Y \times 1$$

$$F_Y = 66 \text{ N}$$

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Object in equilibrium

① Draw a free-body diagram for the object.

② Set up the equations using $F_{net} = 0$.

③ Set up equations using $\tau_{net} = 0$ about any point. Choose a point to set an equation with one unknown if possible.

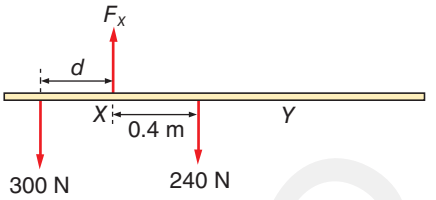
④ Solve the equation(s).

Tip

As the moment provided by F_Y is 0, this equation has one unknown only.

We can also solve for F_Y by using $F_{net} = 0$ along the vertical direction.

(b) Free-body diagram for the tree trunk:



Take moment about X.

Total clockwise moments = total anticlockwise moments

$$240 \times 0.4 = 300 \times d$$

$$d = 0.32 \text{ m}$$

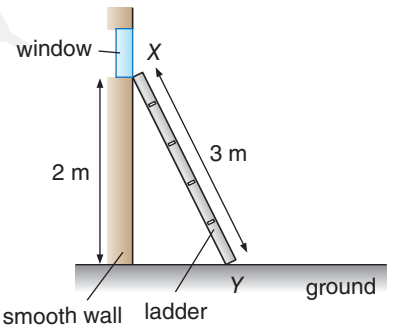
Tip

If the monkey walks too close to the left end, the tree trunk will turn anticlockwise about X. Just before this happens, the tree trunk is about to leave the stone at Y and therefore $F_Y = 0$

Example 5 Forces acting on a ladder

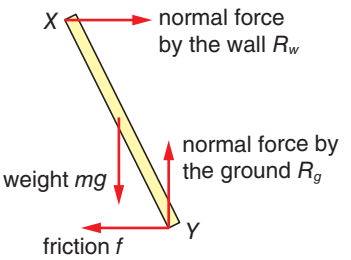
A ladder XY of length 3 m leans against a smooth wall and remains at rest. It just reaches a window 2 m above the ground. The ladder of mass 5 kg is a uniform symmetrical object.

Find the normal force and friction acting on the ladder by the ground.



Solution

The free-body diagram for the ladder is shown on the right.



Consider the vertical direction. Take upwards as positive.

$$F_{\text{net}} = 0$$

$$R_g + (-mg) = 0$$

$$R_g - 5 \times 9.81 = 0$$

$$R_g = 49.05 \text{ N}$$

$$\approx 49.1 \text{ N}$$

Tip

As end Y of the ladder tends to slide to the right, the friction points to the left.

An object moves to lower its c.g. under gravity. Ts may show Ss the following video and ask them to explain what they see.

Another way is taking moment about Y and considering the net force in the horizontal direction.

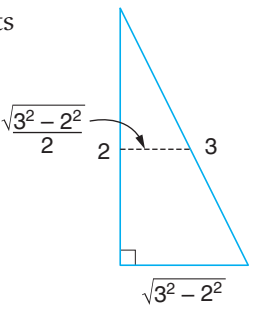
Take moment about X.

Total clockwise moments = total anticlockwise moments

$$f \times 2 + mg \times \frac{\sqrt{3^2 - 2^2}}{2} = R_g \times \sqrt{3^2 - 2^2}$$

$$2f + 5 \times 9.81 \times \frac{\sqrt{5}}{2} = 49.05 \times \sqrt{5}$$

$$f = 27.4 \text{ N}$$



The normal force is 49.1 N (upwards) and the friction is 27.4 N (towards left).

Know more

Iceberg



When an iceberg melts, the position of its c.g. changes and the iceberg is no longer in equilibrium. The moment due to its weight causes it to roll over.

The following video shows an iceberg rolling over.

Video

Quick check 5

Google Forms Microsoft Forms → E-version in OUP Exercise Platform is also available.

1 A plank of weight W is placed on two stones (Fig a). The stones exert upward forces R_1 and R_2 on the plank at points A and B.

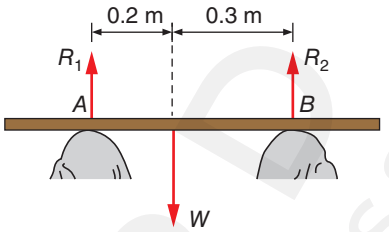


Fig a

(a) Fill each space in the table with **clockwise**, **anticlockwise** or **no moment**.

Force	Direction of moment due to the force	
	about A	about B
W		
R_1		
R_2		

Table a

(b) If W = 500 N, find R_1 and R_2 using Table a.

2 A Chinese scale is still used in some shops (Fig b). The object to be weighed is put on the pan and the mass is moved along the rod until the rod becomes horizontal and stays in equilibrium.

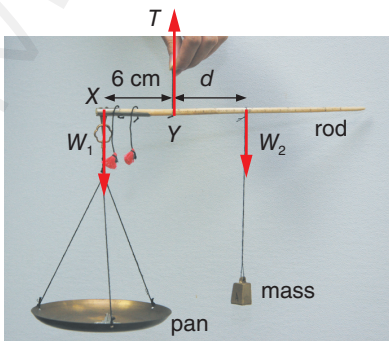


Fig b

(a) Fill each space in the table with **clockwise**, **anticlockwise** or **no moment**.

Force	Direction of moment due to the force	
	about X	about Y
T		
W_1		
W_2		

Table b

(b) If W_1 = 15 N and d = 8 cm, find T and W_2 using Table b.

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Writing practice

Question

In Figure a, point O of the rod is fixed and the rod can rotate about it. An object is tied on one end while a force F acts on the other end to keep the rod remain horizontal. How should the magnitude of F change if an object of larger mass is tied? Explain your answer. (2 marks)

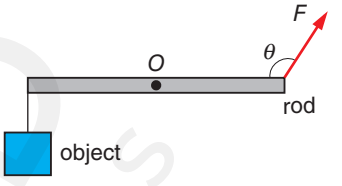


Fig a

Answer

The weight of the object with a larger mass produces a larger moment about O. 1A

To balance this moment, F should increase to produce a larger moment in the opposite direction. 1A

Practise yourself

1 In Figure a, how should the magnitude of F change if the angle θ between F and the rod increases? Explain your answer. (2 marks)

☆ HKDSE 2019 Paper 1A Q7

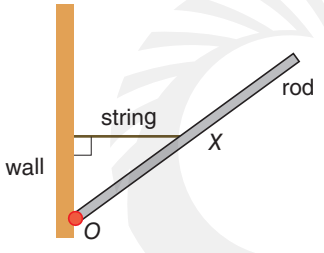
→ This video demonstrates how this question is solved.

Video

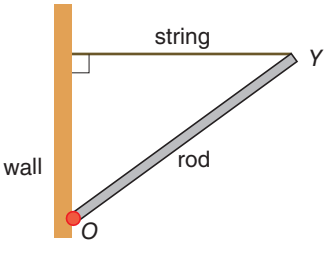
DSE practice 1 | Tying a rod

A rod is attached to a wall at point O and is free to rotate about this point. A string ties the rod to the wall. In which of the following cases is the tension in the string the largest? The angle between the rod and the wall is the same in the four cases.

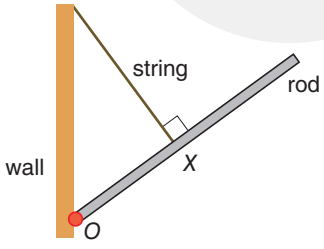
A



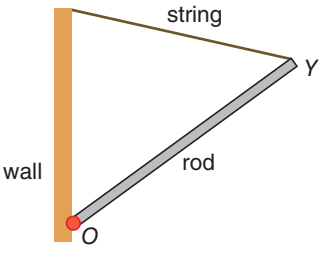
B



C



D



Solution

The tensions should produce the same anticlockwise moment τ about O to balance the clockwise moment due to the weight of the rod in the four cases.

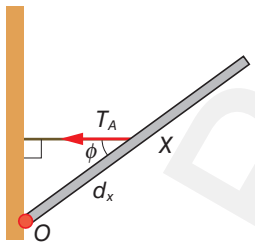
Let $OX = d_X$ and $OY = d_Y$.

Apply $\tau = Fd \sin \theta$.

For A,

$\tau = T_A \times d_X \times \sin \phi$

$T_A = \dfrac{\tau}{d_X \sin \phi}$

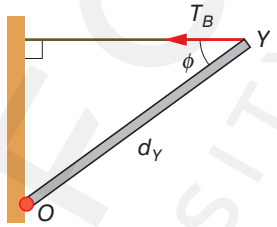


For B,

$\tau = T_B \times d_Y \times \sin \phi$

$T_B = \dfrac{\tau}{d_Y \sin \phi}$

$d_Y > d_X \Rightarrow T_B < T_A$

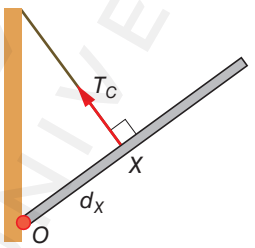


For C,

$\tau = T_C \times d_X$

$T_C = \dfrac{\tau}{d_X}$

$d_X > d_X \sin \phi \Rightarrow T_C < T_A$

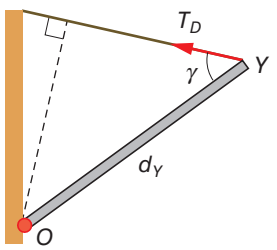


For D,

$\tau = T_D \times (d_Y \sin \gamma)$

$T_D = \dfrac{\tau}{d_Y \sin \gamma}$

$\gamma > \phi \Rightarrow \sin \gamma > \sin \phi \Rightarrow T_D < T_B$



Since $T_B < T_A$, $T_D < T_A$

Common mistake ⚠
Students may not know how to compare T_D with T_A.

$\therefore$ The answer is A.

▶ Revision 5 Q18 (p.222)

☆ HKDSE 2018 Paper 1A Q6

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Direction of force by the ground

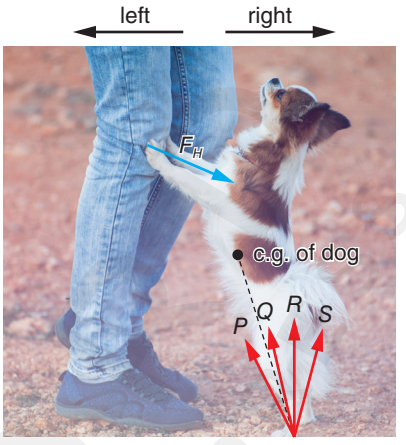
The dog shown in the figure remains stationary. F_H is the force acting on it by the human leg. Which arrow best shows the direction of the force F_G acting on it by the ground?

A P

B Q

C R

D S



Solution

The horizontal component of the net force is zero. Therefore, the horizontal component of F_G must point to the left to balance the horizontal component of F_H which points to the right.

∴ C is incorrect as R does not have a horizontal component.

∴ D is incorrect as the horizontal component of S points to the right.

The net moment about the c.g. of the dog is zero. Therefore, F_G must produce an anticlockwise moment to balance the clockwise moment due to F_H.

∴ A is incorrect as P produces a clockwise moment about the c.g.

∴ The answer is B.

Common mistake ⚠
Students may not be able to apply net moment = 0 to solve the problem.
Note that the moments must be taken about the same point when calculating the net moment.

▶ Revision 5 Q17 (p.222)

Exercise 5.2

★1 A woman remains at rest (Fig a). F is the force acting on her leg by the rock.



Fig a

Which of the following statements about the resultant force acting on her hands by the rock must be correct?

(1) Its vertical component is the same in magnitude as that of F.

(2) Its horizontal component points towards the left.

(3) The moment produced by it about her c.g. is the same in magnitude as that produced by F.

A (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

*2 A rod XZ of 40 cm long is supported by a pivot at Y and is in equilibrium (Fig b).

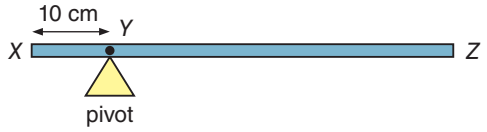


Fig b

What is the ratio of the mass of XY to YZ?

A 1 : 1

B 1 : 3

C 1 : 4

D Cannot be determined

*3 A uniform plank with a weight of 450 N is placed on two trestles X and Y (Fig c) and it is in equilibrium. Find the normal forces acting on the plank by X and Y.

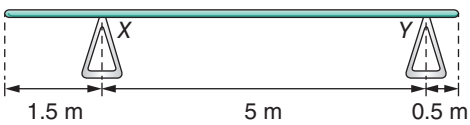


Fig c

*4 A uniform thin rod 1.2 m long is placed on a horizontal surface. The minimum force needed to raise one end of the rod (with the other end kept touching the surface) is 20 N. What is the weight of the rod?

*5 A student supports an umbrella at point X (Fig d) and point Y (Fig e) in turn. The umbrella is in equilibrium in each case.

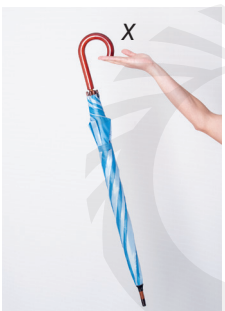


Fig d



Fig e

(a) State the two conditions for an object to be in equilibrium.

(b) Where does the c.g. of the umbrella locate in each case compared to the point of support?

*6 A uniform rod of length 4 m is hinged on a vertical wall at A (Fig f). A string is connected to the rod at B horizontally. The angle between the rod and the wall is 55° and the rod is in equilibrium. If the weight of the rod is 600 N, find the tension in the string.

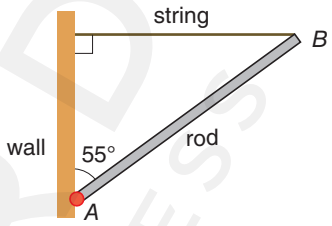


Fig f

*7 A worker stands in the middle of a suspended working platform which is at rest (Fig g). The platform is 3 m wide and its c.g. is at its mid-point. The worker's mass is 70 kg. The platform's mass is 500 kg.

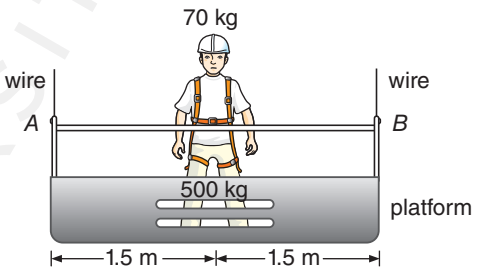


Fig g

(a) Find the tension in each wire suspending the platform.

(b) The worker then walks towards B by 0.5 m.

(i) Explain how the tension in each wire will change without calculation.

(ii) Find the new tension in each wire.

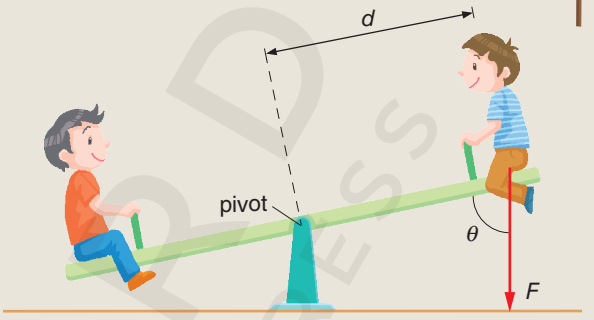
Key concepts 5

1 Moment / Torque

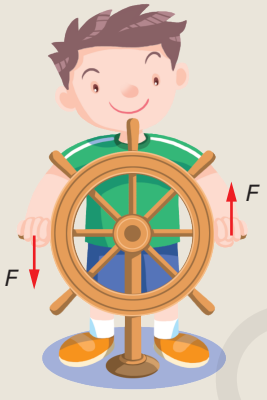
- Turning effect of forces
- Moment about the pivot

  $\tau = Fd_{\perp} = F_{\perp}d = Fd \sin \theta$
- Net moment

  = sum of clockwise moment
  − sum of anticlockwise moment



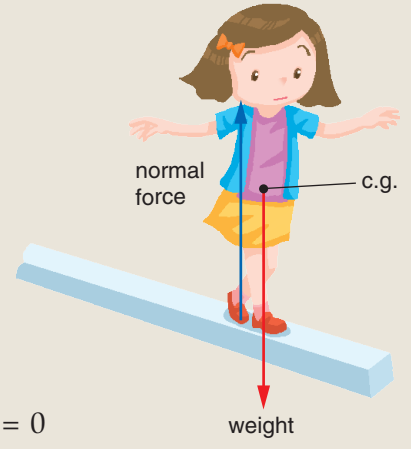
2 Couple



- Two equal and opposite forces apply to the same object and do not act in the same line
- Produces a larger moment than a single force
- Resultant force = 0

3 Centre of gravity (c.g.)

- The point at which the object's weight seems to act.



4 Equilibrium

- Net force = 0
- Net moment about any point = 0

Key terms

1 centre of gravity 重心 p.208
2 couple 力偶 p.201
3 equilibrium 平衡 p.205
4 moment 力矩 p.196
5 pivot 支點 p.199
6 rigid body 剛體 p.196
7 torque 轉矩 p.196

Revision 5

Multiple-choice questions

★1 5.1 A force F acts on a rectangular object as shown (Fig a).

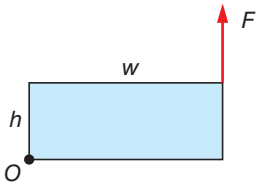


Fig a

What is the moment of F about O?

A Fw
B Fh
C $F\sqrt{w^2+h^2}$
D $\frac{Fw}{h}$

★2 5.2 A nail is being removed from a piece of wood by a hammer (Fig b). When a force of 100 N is applied perpendicularly at the end of the handle of the hammer, the hammer is just able to turn about O and remove the nail. What is the friction f exerted on the nail by the wood? Assume that the hammer exerts a force parallel to f on the nail and the weight of the nail is negligible.

A 1.5 N
B 30 N
C 35 N
D 600 N

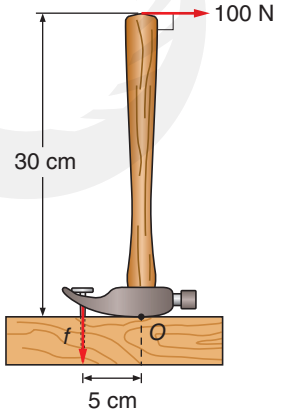


Fig b

★3 5.2 A dog of weight 90 N remains at rest on a horizontal ground (Fig c). It is supported by the ground at points X and Y. The horizontal distances of its c.g. from X and Y are 15 cm and d respectively. The normal force acting on its front legs by the ground at X is 70 N. Find d.

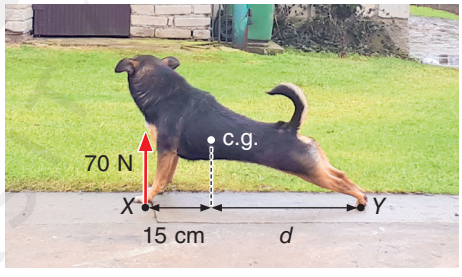


Fig c

A 11.7 cm
B 15 cm
C 20 cm
D 52.5 cm

★4 5.2 Two workers M and N carry a long iron bar on their shoulders (Fig d). M is closer to the c.g. of the iron bar than N.

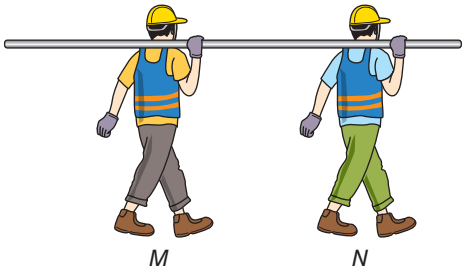


Fig d

If the forces they act on the iron bar point vertically upwards, who provides a larger force?

A M
B N
C They provide forces with the same magnitude.
D Cannot be determined

★5 5.2 The side view of a folding table is shown (Fig e). Its weight is 150 N. A horizontal force F acts at the top end of the table.

What is the minimum value of F needed to make the table topple?

A 10 N
B 12 N
C 14.4 N
D 72 N

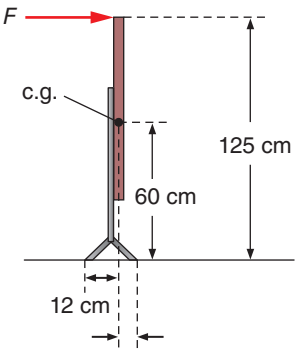


Fig e

★6 5.1 A man applies a force F vertically downwards on a bicycle pedal (Fig f). The pedal is L from the pivot. The crank makes an angle θ with the horizontal. What is the moment of F about the pivot?

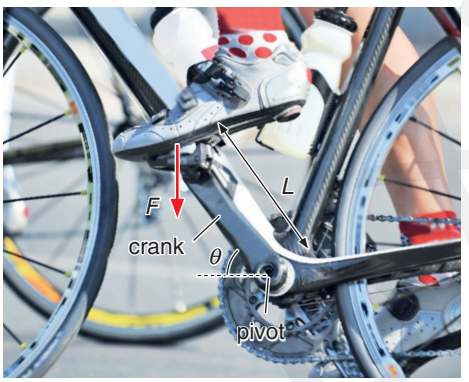


Fig f

A $FL \sin\theta$
B $FL \cos\theta$
C $\dfrac{FL}{\sin\theta}$
D $\dfrac{FL}{\cos\theta}$

★7 5.2 Figure g shows the top view of a uniform square object $ABCD$. There are three horizontal forces acting on it, with the two 3-N forces already shown in the figure.

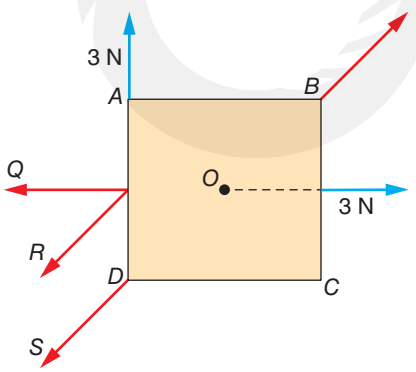


Fig g

If the object is in equilibrium, which arrow best represents the third force?

A P
B Q
C R
D S

★8 5.2 Two identical rods P and Q lean against a smooth wall at different angles (Fig h). They remain at rest. F_P and F_Q are respectively the normal force acting on P and Q by the wall.

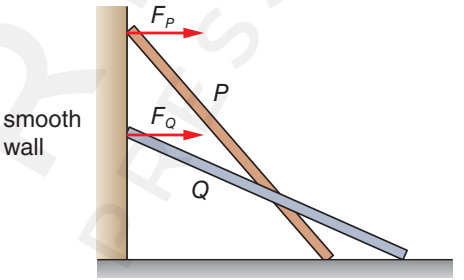


Fig h

Which of the following is correct?

A $F_P > F_Q$
B $F_P = F_Q$
C $F_P < F_Q$
D The relationship between F_P and F_Q cannot be determined.

(For Q9–10). A uniform gangplank of weight 200 N is hinged smoothly to a ferry at point O. It is held stationary by a string attached to its c.g. (Fig i).

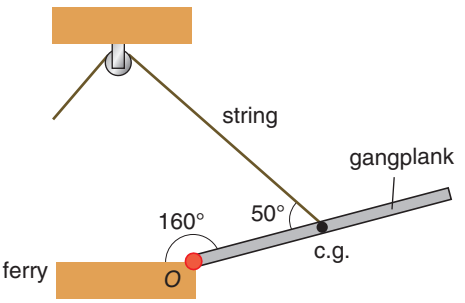


Fig i

★9 5.2 What is the tension in the string?

A 100 N
B 153 N
C 200 N
D 245 N

★10 5.2 In which direction does the force acting on the gangplank by the ferry at O point (Fig j)?

A P
B Q
C R
D S

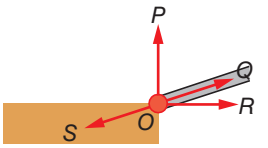


Fig j

*11 5.2 Sam of weight 500 N stands on a diving board which is fixed by two supports at points *X* and *Y* as shown (Fig k). The weight of the board is 200 N and the c.g. of the board is 1.2 m from *Y*.

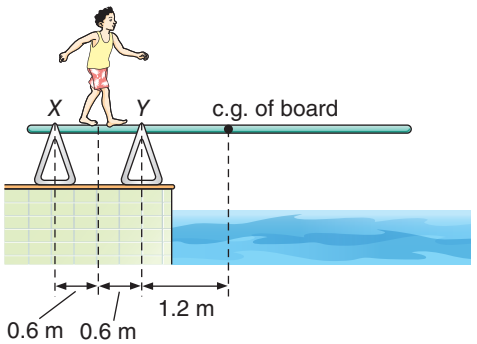


Fig k

Sam walks from the position shown in Figure k to *Y*. Which of the following best describes the change in the force acting on the board by the support at *X*?

A Decreases in magnitude and remains upwards

B Decreases in magnitude and remains downwards

C Decreases in magnitude and eventually changes from upwards to downwards

D Increases in magnitude and remains downwards

*12 5.2 A pot holder is hung on a vertical fence (Fig l). *P* and *Q* are the two contact points between the holder and the fence. The force acting on the holder at *Q* by the fence is along the horizontal direction. The common c.g. of the pot and the holder is at *C*. Which arrow best shows the direction of the force acting on the holder at *P* by the fence?

A *W*

B *X*

C *Y*

D *Z*

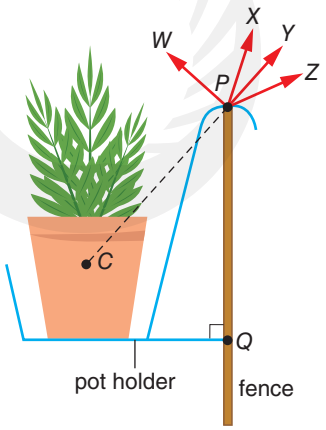


Fig l

*13 5.2 A man is practicing human flag in a park (Fig m). He holds points *P* and *Q* on a vertical ladder. These two points are 1 m apart. His weight is 700 N and his c.g. is 0.8 m from the ladder.

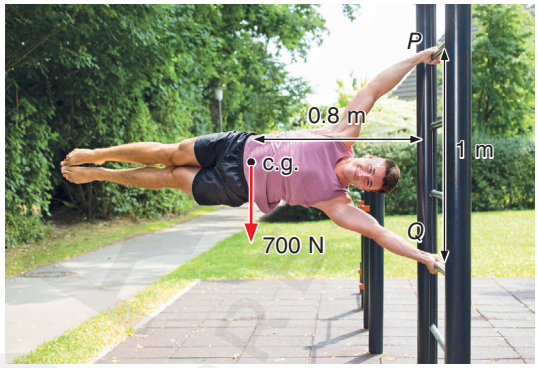


Fig m

What is the horizontal component of the force acting on him at *Q* by the ladder?

A 280 N (towards left)

B 280 N (towards right)

C 560 N (towards left)

D 560 N (towards right)

**14 5.2 A plank *PQ* leans against a smooth wall (Fig n).

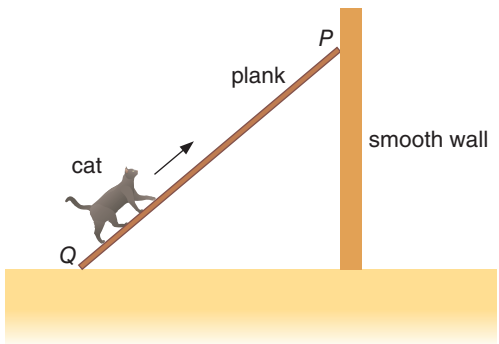


Fig n

How do the normal force *N* and friction *f* acting on the plank at *Q* change when a cat walks at a constant velocity from *Q* to *P*? The plank remains at rest.

	N	*f*
A	increases	increases
B	increases	decreases
C	remains unchanged	increases
D	remains unchanged	decreases

▶ Refer p.212

15 **HKDSE 2012 Paper 1A Q6**
5.2

A uniform gangplank PQ of a ferry smoothly hinged at end P initially rests horizontally on the pier. The gangplank has mass M and length 2 m. It is raised by a man on the ferry using a light rope passing a smooth fixed light pulley and connecting to R on the gangplank as shown. R is 1.5 m from end P. Which of the following correctly describes the force required to raise the gangplank steadily?

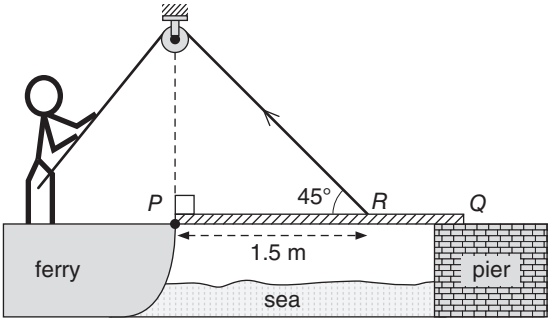


Fig o

	Initial force required to raise the gangplank when it is horizontal	*Subsequent* force required to raise the gangplank
A	$0.67Mg$	greater than $0.67Mg$
B	$0.67Mg$	smaller than $0.67Mg$
C	$0.94Mg$	greater than $0.94Mg$
D (41%)	$0.94Mg$	smaller than $0.94Mg$

16 **HKDSE 2016 Paper 1A Q11**
5.2

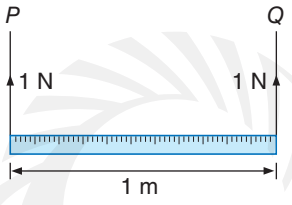


Fig p

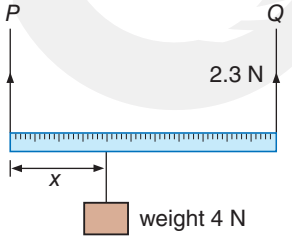


Fig q

A uniform metre rule is supported by vertical wires P and Q and remains at rest horizontally as shown in Figure p. The tension in each wire is 1 N. When a weight of 4 N is hung from the metre rule at a certain position as shown in Figure q, the tension in Q becomes 2.3 N while the metre rule remains horizontal. Find the distance x shown.

A 32.5 cm (49%)

B 57.5 cm

C 67.5 cm

D Answer cannot be found as the tension in P is not known.

17 **HKDSE 2018 Paper 1A Q6**
5.2

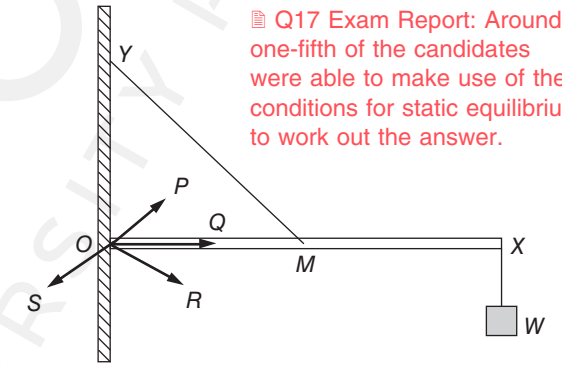


Fig r

Q17 Exam Report: Around one-fifth of the candidates were able to make use of the conditions for static equilibrium to work out the answer.

A uniform light rigid rod OX is hinged smoothly to a wall at one end O. Its mid-point M is connected by a light inextensible string to a point Y directly above O while a weight W is suspended from the other end X of the rod as shown. Rod OX remains horizontal. The reaction force acting on the rod due to the wall is along the direction

A OP. (30%)

B OQ. (39%)

C OR. (21%)

D OS. (10%)

18 **HKDSE 2019 Paper 1A Q7**
5.2

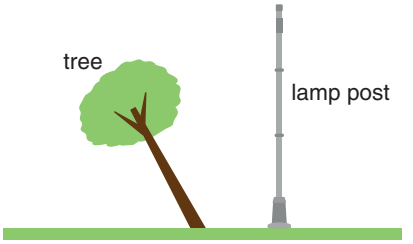
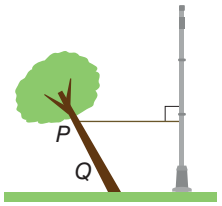


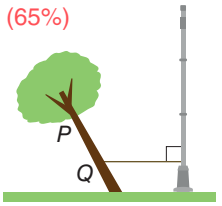
Fig s

A tree blown by strong wind is found leaning to one side. In order to give the tree some support, a rope is wrapped around its trunk and tied to a fixed lamp post nearby. In which of the following arrangements would the rope have the highest chance of breaking?

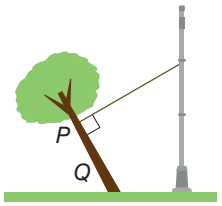
A



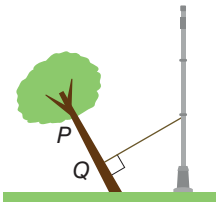
B (65%)



C



D



19 5.2 **HKDSE 2020 Paper 1A Q7**

Figure t shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.

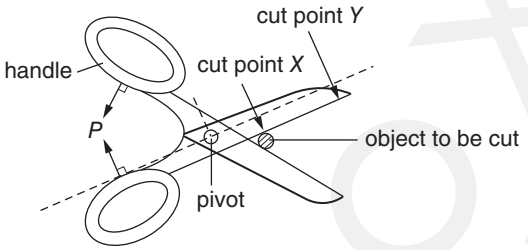
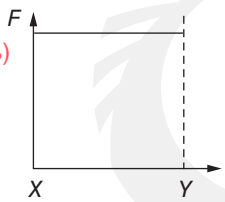


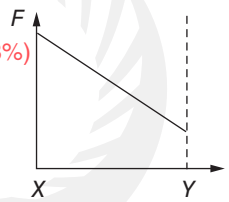
Fig t

Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed?

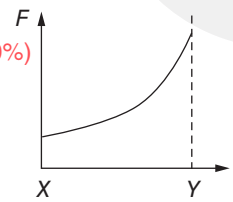
A (8%)



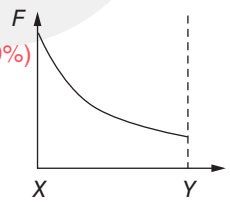
B (43%)



C (20%)



D (29%)



Q19 Exam Report: While over 70% of the candidates indicated knowledge that the force is larger at cut point X, less than 30% of them fully understood how the force varies when the pair of scissors is closed.

Conventional questions

★**20** 5.2 A man holds a barbell still (Fig u). The weight of the barbell is 700 N and its c.g. is at O. The forces that the man's hands exert on the barbell are F_R and F_L.

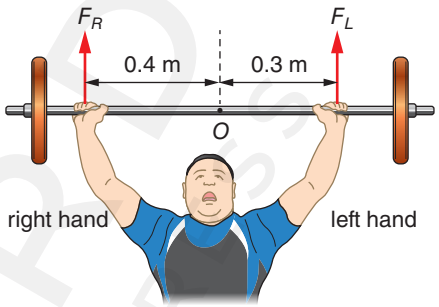


Fig u

(a) Find F_R and F_L. (4 marks)

(b) Describe how F_R and F_L will change if the man's right hand moves away from O. (2 marks)

★**21** 5.2 Tom hangs a rectangular board of weight 10 N on a wall using two pieces of string as shown (Fig v). The strings are each 0.8 m long. The c.g. of the board is at its centre C.

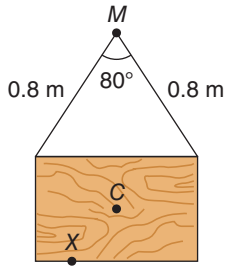


Fig v

(a) Draw a free-body diagram for the board. (2 marks)

(b) Find the tension in each string. (2 marks)

(c) When Tom hangs a heavy object at point X on the board, one of the strings breaks. The object then falls onto the ground and the board is suspended by the remaining string.

(i) If the two strings are equally strong, which string has broken? Explain briefly. (3 marks)

(ii) Draw a diagram to show how the board is suspended by the remaining string. (1 mark)

★22 5.2 A man bends his back to lift a box (Fig w). Suppose he remains stationary at the moment shown.



Fig w

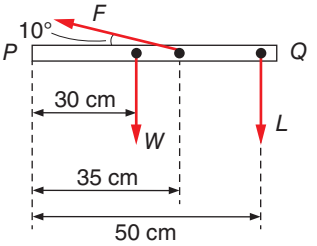


Fig x

His spine is represented by a straight rod PQ pivoted at P (Fig x). Four forces W, F, L and R act on it. W is the weight of his upper body. F is the resultant force by the back muscles. L is the load due to the weight of the hands and the box. R, which is not shown in the figure, is the force acting at P by the pivot.

(a) Sketch the direction of R in Figure x. (1 mark)

(b) Take W = 500 N and L = 100 N.

(i) Find the magnitude of F. (2 marks)

(ii) Find the component of R along the spine. (2 marks)

(iii) Under normal conditions (Fig y), the force acting at P along the spine is equal to the weight of the upper body. Explain why lifting the load in the way shown in Figure w may cause injury to the spine. (1 mark)



Fig y

★23 Misc A woman uses a carrying pole to carry loads (Fig z). The mass of the pole is negligible compared with the loads. Suppose that the weight of the load at each end is 200 N. The pole is horizontal and the points where the loads are hung are equidistant from the point supported by the woman's shoulder.



Fig z

(a) Do the forces acting on the pole by the loads form a couple? Explain briefly. (2 marks)

(b) Find the force acting on the woman's shoulder by the pole. Assume that the woman's hands do not apply any force on the load. (2 marks)

(c) Would it be easier or harder for the woman to carry the loads if both loads are placed closer to the woman and are still equidistant from her? (1 mark)

(d) How would the woman change the distances of the loads from her if the loads are of different masses? (1 mark)

(e) Would it be easier or harder for the woman to carry the loads if the two loads are put together as shown in Figure aa? Explain briefly. (3 marks)

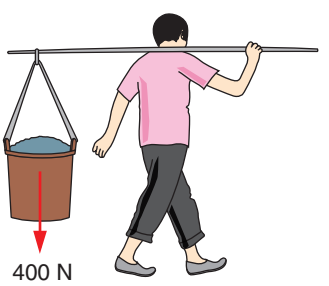


Fig aa

★24 Misc Read the following passage about **pet's bowls** and answer the questions that follow.

Pet owners usually use special bowls to hold their pets' food (Fig ab).



Fig ab

This kind of bowl has a low centre of gravity and a large base radius compared with its height. Besides, the side of the bowl is not vertical, but slanted. These features reduce the chance of it toppling. We shall see below that increasing the base radius alone can help stabilize the bowl.

Let P be the centre of gravity of the bowl, Q be the point on the ground vertically below P and W be the weight of the bowl. Suppose a force F is exerted on the bowl so that the bowl tends to topple clockwise about T (Fig ac). Consider the moment τ of W about T:

$$\tau = W \times QT$$

This moment is anticlockwise and therefore tends to balance the moment by F. It can be seen that increasing QT (i.e. the base radius) helps stabilize the bowl.

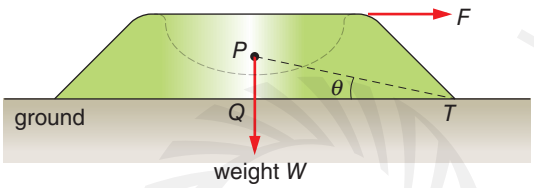


Fig ac

(a) A dog licks point S of its bowl after a meal (Fig ad). It exerts a force of 3 N up the slanted surface. The base radius of the bowl is 9 cm. The slanted surface makes an angle of 45° to the ground. The bowl is 5 cm high.

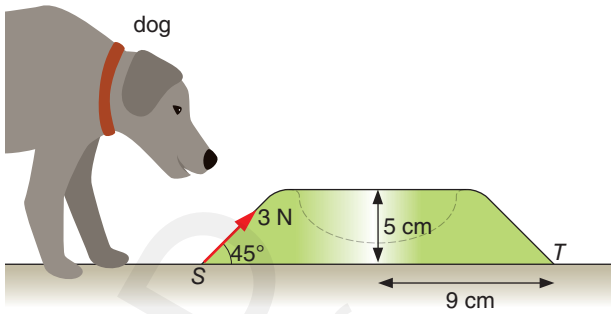


Fig ad

(i) Find the moment of the 3-N force about T. (2 marks)

(ii) Show that in this case, decreasing the slanting angle helps stabilize the bowl. (2 marks)

(b) The dog accidentally kicks point S with a horizontal force 10 N. Find the moment of this force about T. (1 mark)

★25 5.2 A robot of weight 50 N is put on a uniform plank suspended horizontally by two strings A and B (Fig ae). The weight of the plank is 150 N. The tension in string A is 110 N.

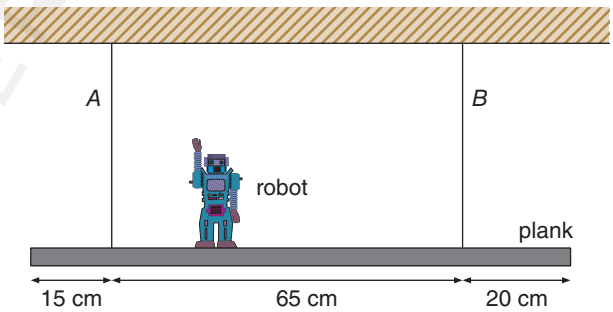


Fig ae

(a) Find the tension in string B. (2 marks)

(b) Find the distance between the robot and string A. (2 marks)

(c) String A will break if the tension in it exceeds 115 N.

(i) Find the minimum distance between the robot and string A without breaking the string. (2 marks)

(ii) Can the robot be placed on the left of string A without breaking the string? (1 mark)

**26 5.2 A student uses the following set-up to study the moment (Fig af). A uniform rod of 40 cm long is hung by a string at *X*. A weight of 10 N is hung at one end of the rod. The student applies a downward force *F* on the rod with a spring balance to keep the rod horizontal.

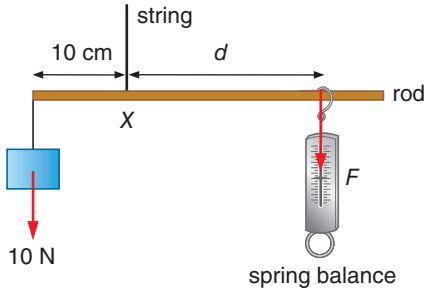


Fig af

The student varies the distance *d* of the spring balance from *X* and reads the corresponding value of *F* from the spring balance. His results are shown in Table a.

***d* / cm**	25	20	15	10	5
***F* / N**	3.1	3.7	5.0	7.4	14.9

Table a

(a) Does the tension in the string from which the rod is hanging remain constant in the experiment? Explain briefly. (2 marks)

(b) Plot a straight-line graph relating *F* and *d*. (3 marks)

(c) Find the slope of the graph in (b). What is its physical meaning? (2 marks)

(d) Consider the moment of *F* and the moment due to the 10-N weight about *X*.

(i) The magnitudes of the two moments are not equal. Explain the difference. (2 marks)

(ii) Hence suggest a modification to the experiment so that the two moments become equal in magnitude. (1 mark)

▶ Refer p.211

☆ Physics A Unit G481/01

27 5.2 OCR GCE AS Jan 2012 Q4

(a) Define moment of a force. (1 mark)

(b) State the two conditions that apply when an object is in equilibrium. (2 marks)

(c) Figure ag is a diagram of a human arm lifting an object.

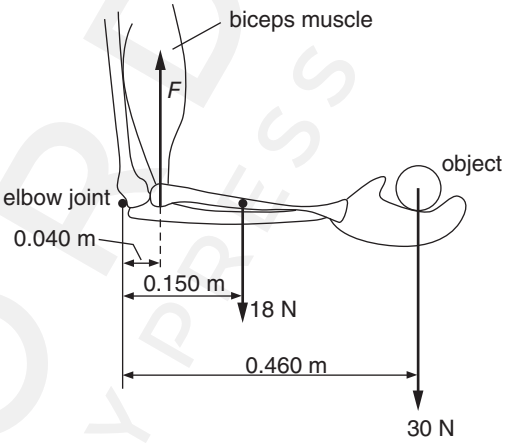


Fig ag

The lower arm is horizontal and its centre of gravity is 0.150 m from the elbow joint. The weight of the lower arm is 18 N. The biceps muscle exerts a vertical force *F* on the arm. The horizontal distance between the elbow joint and the point of attachment of the muscle to the lower arm bone is 0.040 m. The weight of the object held in the hand is 30 N and its centre of gravity is 0.460 m from the elbow joint. The arm is in equilibrium.

(i) Define centre of gravity. (1 mark)

(ii) Calculate the total clockwise moment about the elbow joint. (2 marks)

(iii) As the lower arm is moved away from the body, the force *F* exerted by the biceps muscles acts at an angle θ to the vertical as shown in Figure ah. The lower arm remains horizontal and in equilibrium. Describe and explain what happens to each of the following quantities as the angle θ is increased.

(1) The anticlockwise moment about the elbow joint

(2) The magnitude of the force *F* (3 marks)

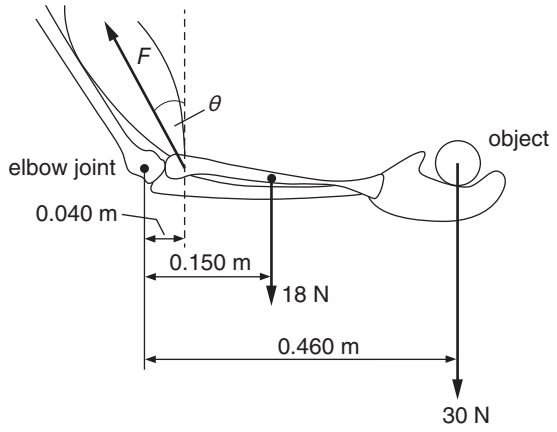


Fig ah

28 HKDSE 2018 Paper 1B Q5

Misc

Figure ai shows the use of a beam balance to measure the mass of a load. Without the load and the counter-weight, the beam with a scale pan at its left end is balanced and remains horizontal. (g = 9.81 m s^{-2})

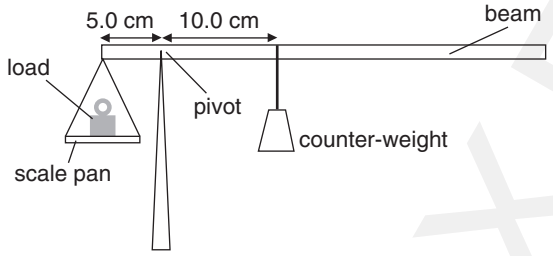


Fig ai

(a) A load is placed on the pan which is 5.0 cm from the pivot. The set-up is in equilibrium when the counter-weight of mass 50 g is 10.0 cm away from the pivot as shown.

(i) Find the mass of the load. (2 marks)

(ii) If the reading of the counter-weight's position on the beam carries an uncertainty of ±0.1 cm, find the maximum error that corresponds to the result found in (a)(i). (2 marks)

(b) The weight of an identical load is now measured by a spring balance calibrated in newtons. What is the balance's reading? (1 mark)

(c) The beam balance set-up in Figure ai and the spring balance with the load in (b) are both brought into a lift.

(i) The measurements are then repeated inside the lift which accelerates upwards uniformly. State the change, if any, in the measurements respectively. (2 marks)

counter-weight position on beam balance	spring balance reading

(ii) A student claims that if the lift falls freely, the beam balance can still be used to measure the mass of the load. Explain whether the claim is correct or not. (2 marks)

Q28 Exam Report: In (a)(i), most candidates correctly considered the balance of moments in the system. However, a few made mistakes in the conversion of units (like 1 cm = 0.001 m or 50 g = 0.5 kg). (a)(ii) was new to most candidates. Many considered the maximum and minimum values and found their difference instead of using percentage error. Some candidates overlooked that only the error associated with the counter-weight's position was considered or used ±0.05 cm in their calculations. In (b), most were able to use $W = mg$ to find the correct value. In (c), only some candidates were able to correctly state the counter-weight position on the beam balance. Most were able to indicate that there was an increase of spring balance reading. Candidates' performance in (c)(ii) was poor. Very few were able to precisely explain why the beam balance failed to work in a free falling lift. Some of them wrongly thought that the beam balance reading would decrease.

Know more

The working principle of a beam balance (p.206)

The two arms of a beam balance are of equal length.

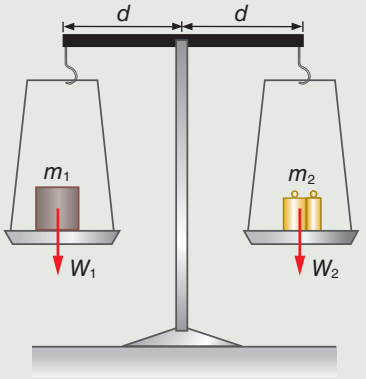
When the balance is in equilibrium,

$W_1 d = W_2 d$

$m_1 g d = m_2 g d$

$m_1 = m_2$

∴ When a beam balance is in equilibrium, the objects on the two sides have the same mass.



Know more

Position of c.g. (p.210)

Consider an object which is only supported at one point from below. When the c.g. is slightly displaced sideways, the moment due to the weight will make the object topple (Fig a). Therefore, it will be hard to keep its balance.

However, if the c.g. of an object is below the supporting point, even when the c.g. is displaced sideways, the moment due to the weight will turn the object back to its equilibrium position (Fig b). It is easy for such an object to restore its balance.

The pen and the dry cell on p.205 can restore their equilibrium position even when the pen is tapped lightly. This can be explained by the position of the c.g. (Fig c).

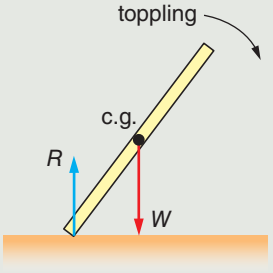


Fig a

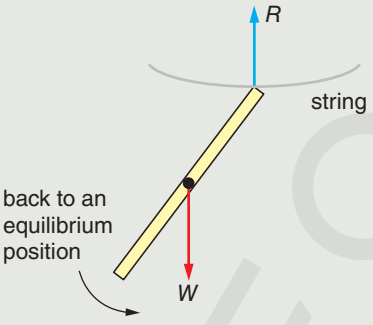


Fig b

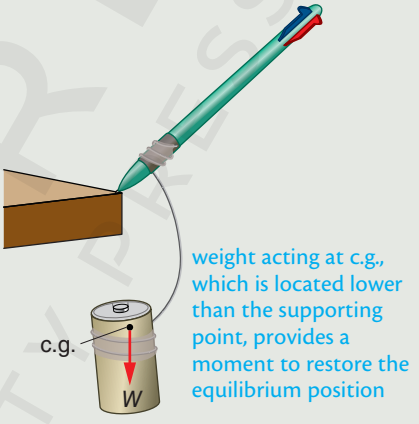


Fig c

If an object is supported at many points, the region enclosed by these points is the base of support. The object is stable as its c.g. is above the base of support (Fig d).

If this object is tilted and becomes supported at only one point, its weight will produce a moment about that point of support and turns the object. The object **may return to its stable position or topple, depending on the direction of moment** due to the weight (Fig e).

R_1 and R_2 are not necessarily equal. However, the following conditions must be satisfied:
1 $R_1 + R_2 = W$ (net force = 0)
2 Net moment about each point of support = 0

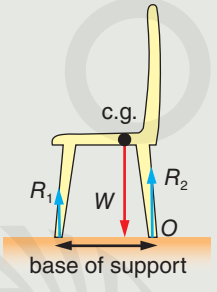


Fig d

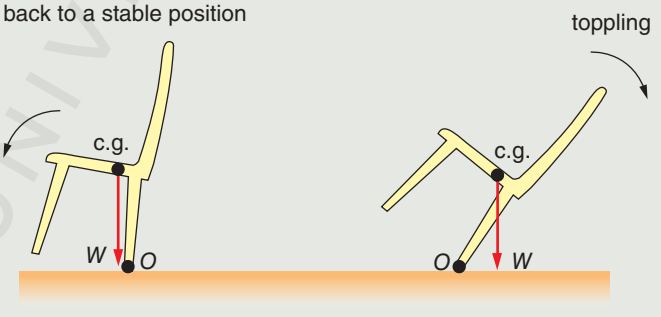


Fig e

An object will be less likely to topple if it has a wider base of support or a lower c.g. Practically, widening the base of support and lowering the c.g. are two common methods used to lower the risk of toppling (Fig f and g).



Fig f *Extending legs are used to widen the base of support.*



Fig g *Khalifa tower, the tallest building in the world, has a wide base and a low c.g.*